

An Introduction to Gaussian Processes (GP)

Le Yang, Joe Chen and Jeremy Waston

Department of ECE, University of Canterbury

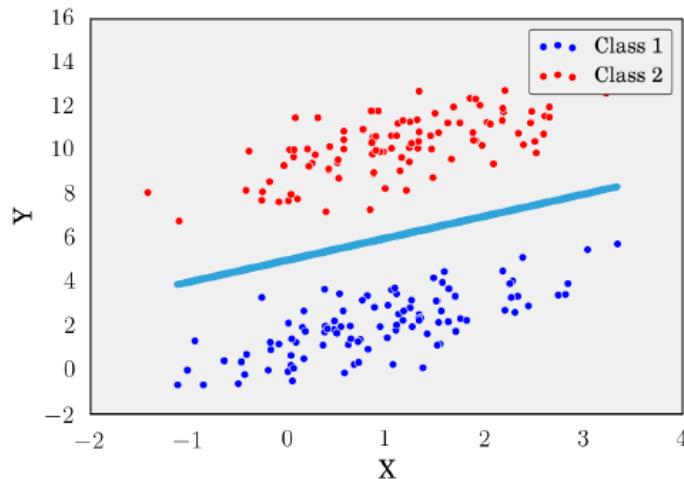
Outline

- Supervised learning and regression
- Fundamentals of Gaussian Processes (GP)
- Mean function, covariance function and hyperparameters in GP
- GP for nonlinear regression
- GP in spatiotemporal modeling
- References

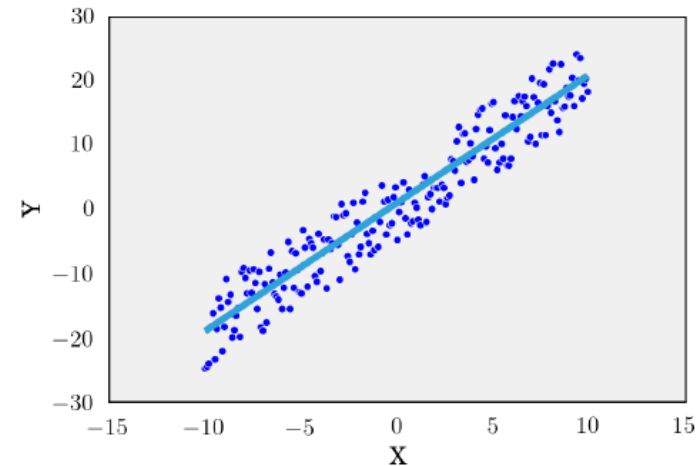
Supervised Learning

- Supervised learning
 - Establish (generally non-linear) curve(s) to
 - Build decision boundaries (classification)
 - Fit through labeled data (regression)

Classification Problem



Regression Problem



Supervised Regression

- Formulation of a regression problem
 - Given a set of N input-output pairs (**labeled training data**) $\{\mathbf{x}_n, y_n\}_{n=1}^N$
 - $\mathbf{x}_n$: input
 - y_n : output
 - Find a nonlinear (multivariate) function $y = f(\mathbf{x})$ that can
 - Describe the training data well (**no underfitting**)
 - Generalize to inputs other than $\mathbf{x}_n$ well (**no overfitting**)
 - Predictive/generative modeling

Fundamentals of Gaussian Processes (1)

- Definition
 - A Gaussian Processes (GP) is a collection of random variables with **the joint distribution of any subset being Gaussian**
- To specify a (multivariate) Gaussian distribution, we only need to provide
 - Mean and covariance
- Similarly, to specify a GP, we need
 - Mean function
 - Covariance function

Fundamentals of Gaussian Processes (2)

- Impose GP prior on the function $y = f(\mathbf{x})$ to be estimated
 - **Any** subset of function values $y_i = f(\mathbf{x}_i)$ is Gaussian
 - $p(y_1, y_2, \dots, y_M)$ is Gaussian
 - $M = 1, 2, 3, \dots$ is an **arbitrary** positive integer
- What are the mean and covariance for $p(y_1, y_2, \dots, y_M)$?
 - Note that the inputs for y_i are different
 - Cannot use **fixed** mean and covariance
 - Mean and covariance should depend on the input $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M$

Fundamentals of Gaussian Processes (3)

- Mean function $m(\mathbf{x})$
 - Specify the mean vector for $p(y_1, y_2, \dots, y_M)$
 - $\boldsymbol{\mu}_{M \times 1} = \begin{bmatrix} m(\mathbf{x}_1) \\ \vdots \\ m(\mathbf{x}_M) \end{bmatrix}$
- Covariance function $k(\mathbf{x}, \mathbf{x}')$
 - Specify the covariance matrix for $p(y_1, y_2, \dots, y_M)$
 - $\boldsymbol{\Sigma}_{M \times M} = \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & \cdots & k(\mathbf{x}_1, \mathbf{x}_M) \\ \vdots & \ddots & \vdots \\ k(\mathbf{x}_M, \mathbf{x}_1) & \cdots & k(\mathbf{x}_M, \mathbf{x}_M) \end{bmatrix}$

Mean Function and Covariance Function in GP (1)

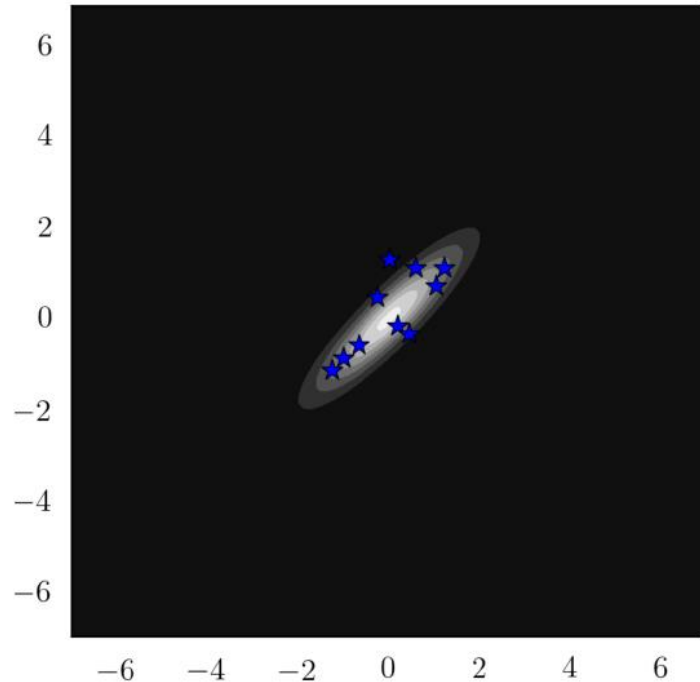
- Mean function $m(\mathbf{x})$ captures the **trend** in the function $f(\mathbf{x})$
 - $m(\mathbf{x}) = 0$ is commonly used and in many cases, a good choice
 - $m(\mathbf{x}) = \sum_{j=1}^J a_j \varphi_j(\mathbf{x})$, $\varphi_j(\mathbf{x})$ is a linear/nonlinear function of $\mathbf{x}$ (surrogate 😊)
- Covariance function $k(\mathbf{x}, \mathbf{x}')$
 - Specifies the **variance** of the function value when $\mathbf{x} = \mathbf{x}'$
 - Specifies the **correlation** between function values $f(\mathbf{x})$ and $f(\mathbf{x}')$
- $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$: $f(\mathbf{x})$ has a GP prior with mean function $m(\mathbf{x})$ and covariance function $k(\mathbf{x}, \mathbf{x}')$

Mean Function and Covariance Function in GP (2)

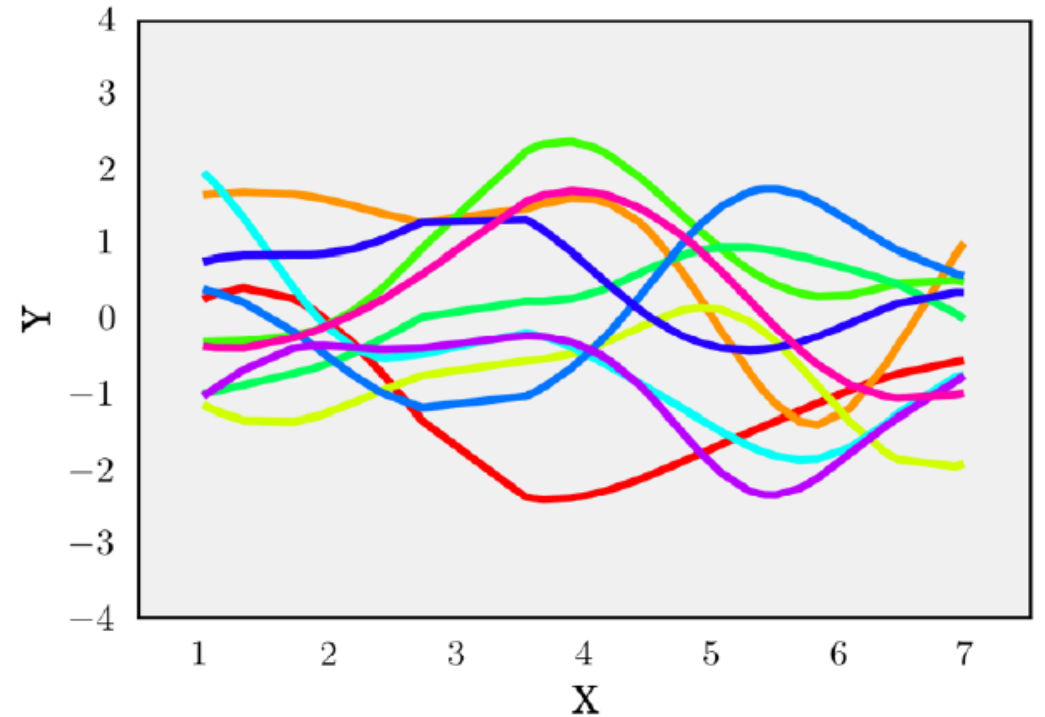
- Squared exponential (SE): a commonly used covariance function
 - $k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2l^2}\right)$
 - $k(\mathbf{x}, \mathbf{x}) = \sigma_f^2$
 - σ_f^2 : *a priori* variance of function values $f(\mathbf{x})$
 - l : characteristic length
 - Smoothness of the function $f(\mathbf{x})$ to be modeled
- What if the training data has noise in y_n ?
 - $k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2l^2}\right) + \sigma_v^2 \delta(\mathbf{x} - \mathbf{x}')$
 - σ_v^2 : noise variance
 - $\delta(\cdot)$: Dirac delta function

Mean Function and Covariance Function in GP (3)

- Sampling from $z \sim N(\mu, \Sigma)$ v.s. sampling from $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$



$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}$$



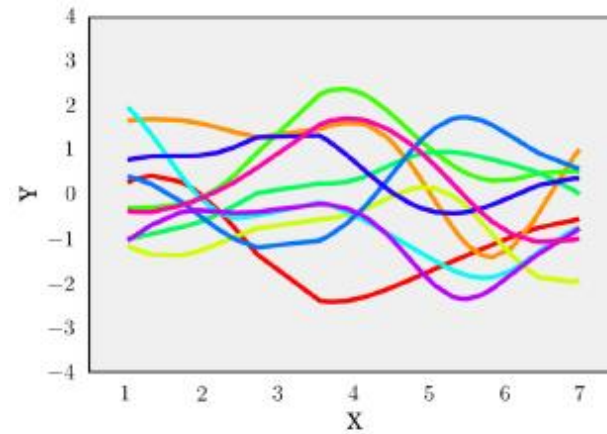
$$m(x) = 0, k(x, x') = \exp(-||x - x'||^2) + \delta(x - x')$$

Hyperparameters in GP (1)

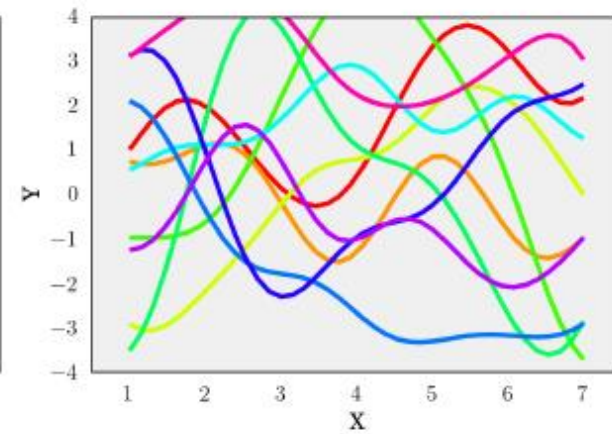
- Hyperparameters in GP
 - *Unknown* parameters in the mean function $m(\mathbf{x})$ and covariance function $k(\mathbf{x}, \mathbf{x}')$
- Example
 - $m(\mathbf{x}) = \sum_{j=1}^J a_j \varphi_j(\mathbf{x})$
 - $k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2l^2}\right) + \sigma_v^2 \delta(\mathbf{x} - \mathbf{x}')$
 - $J + 3$ hyperparameters: $\{a_1, a_2, \dots, a_J, l, \sigma_f^2, \sigma_v^2\}$

Hyperparameters in GP (2)

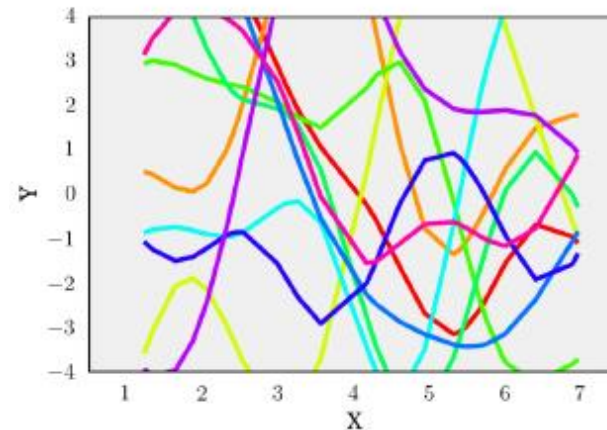
- Sampling from $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$
 - $m(x) = 0$
 - $k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2l^2}\right) + \sigma_v^2 \delta(x - x')$
 - $l = 1$
 - $\sigma_v = 0$: no measurement noise
 - (a) $\sigma_f = 1$
 - (b) $\sigma_f = 2$
 - (c) $\sigma_f = 3$
 - (d) $\sigma_f = 0.5$
- σ_f specifies the 'signal' energy



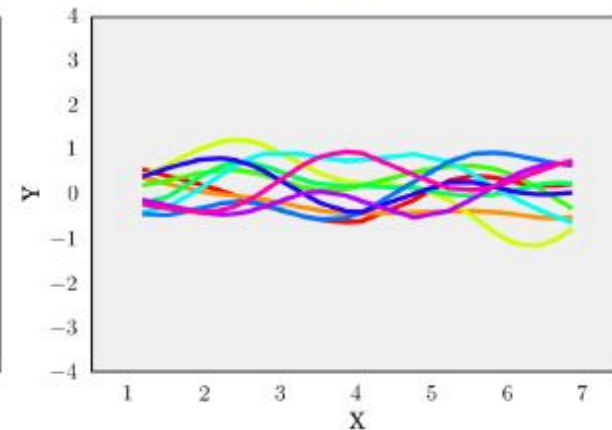
(a)



(b)



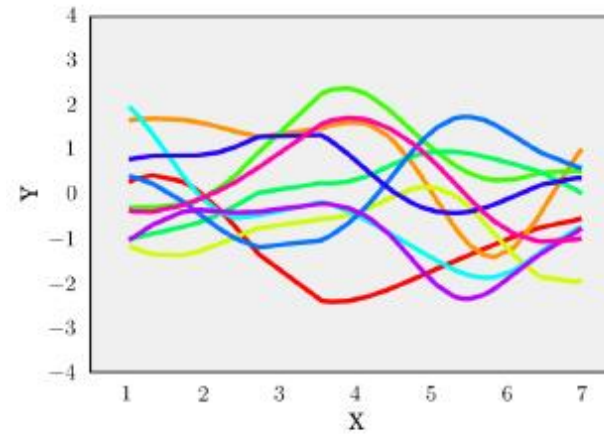
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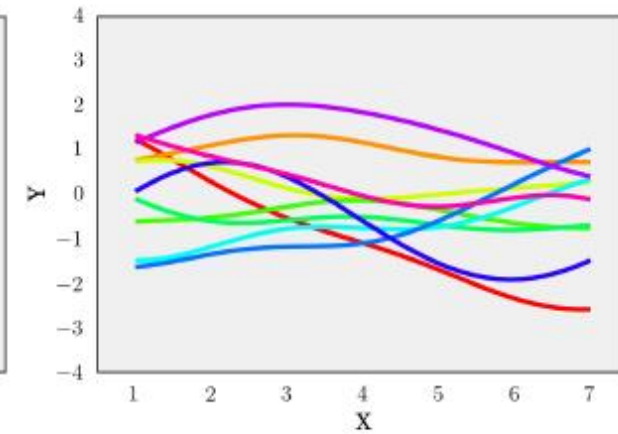
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Hyperparameters in GP (3)

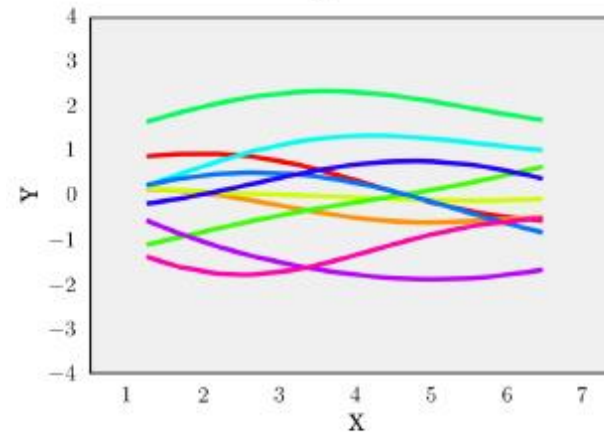
- Sampling from $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$
 - $m(x) = 0$
 - $k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2l^2}\right) + \sigma_v^2 \delta(x - x')$
 - $\sigma_v = 0$: no measurement noise
 - $\sigma_f = 1$
 - (a) $l = 1$
 - (b) $l = 3$
 - (c) $l = 5$
 - (d) $l = 0.5$
- l specifies the smoothness of the function



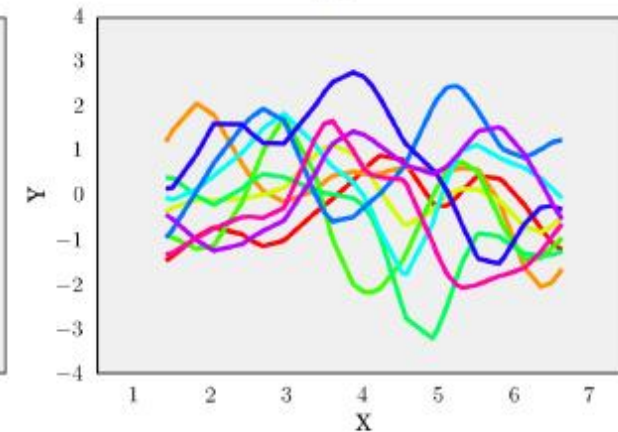
(a)



(b)



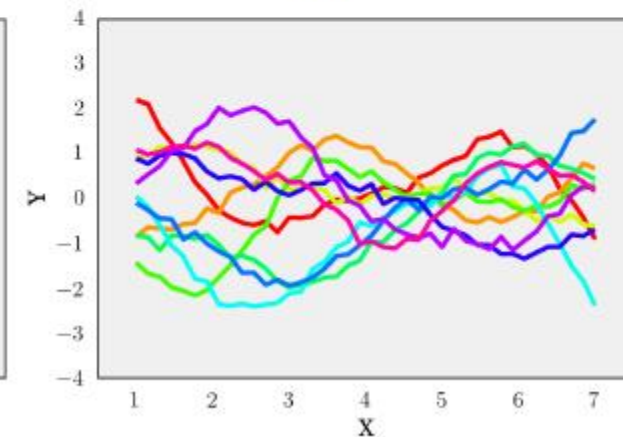
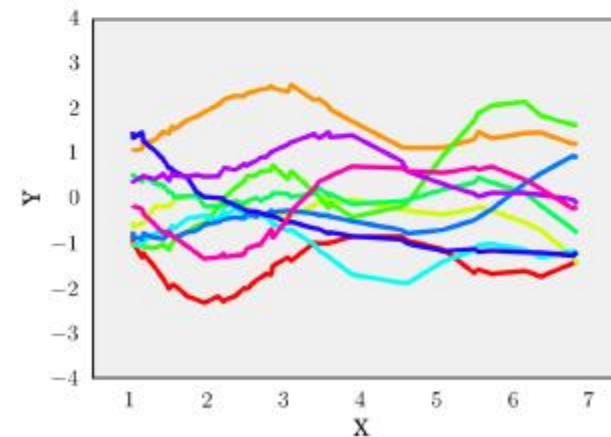
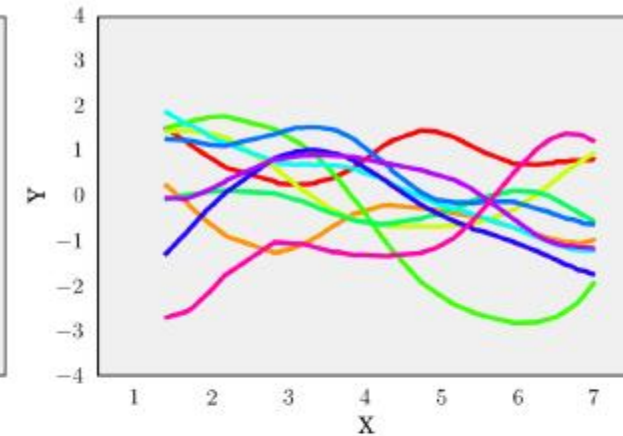
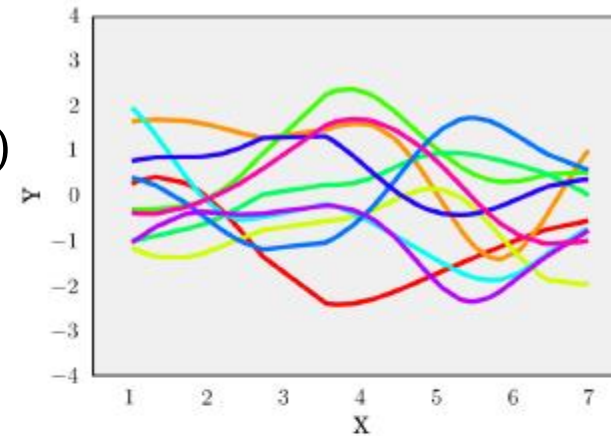
(c)



(d)

Hyperparameters in GP (4)

- Sampling from $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$
 - $m(x) = 0$
 - $k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2l^2}\right) + \sigma_v^2 \delta(x - x')$
 - $l = 1$
 - $\sigma_f = 1$
 - (a) $\sigma_v = 0$
 - (b) $\sigma_v = 0.01$
 - (c) $\sigma_v = 0.05$
 - (d) $\sigma_v = 0.1$
- σ_v specifies the SNR



Hyperparameters in GP (5)

- Maximum likelihood (ML) estimation of GP hyperparameters

- $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$

- The training data satisfy

$$p(y_1, y_2, \dots, y_N) = N \left(\begin{bmatrix} m(\mathbf{x}_1) \\ \vdots \\ m(\mathbf{x}_N) \end{bmatrix}, \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & \cdots & k(\mathbf{x}_1, \mathbf{x}_N) \\ \vdots & \ddots & \vdots \\ k(\mathbf{x}_N, \mathbf{x}_1) & \cdots & k(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix} \right)$$

- Find the hyperparameters through solving **via conjugate gradient**

$$\max \log(p(y_1, y_2, \dots, y_N))$$

GP for Nonlinear Regression (1)

- Problem setup
 - Given a set of N training data points $\{\mathbf{x}_n, y_n\}_{n=1}^N$
 - The function to be modeled, $f(\mathbf{x})$, has a GP prior $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$
 - GP hyperparameters have been found through e.g., ML estimation
 - Data-driven approach
- We are interested to find the posterior
$$p(y^* | \mathbf{x}^*, \{\mathbf{x}_n, y_n\}_{n=1}^N)$$
 - $y^* = f(\mathbf{x}^*)$: the function value at an input $\mathbf{x}^*$
 - ‘Predictive’ probability of y^*
 - Modeling of mapping from $\mathbf{x}^*$ to y^*

GP for Nonlinear Regression (2)

- By Bayes theorem

$$p(y^* | \mathbf{x}^*, \{\mathbf{x}_n, y_n\}_{n=1}^N) = \frac{p(\{\mathbf{x}^*, y^*\}, \{\mathbf{x}_n, y_n\}_{n=1}^N)}{p(\{\mathbf{x}_n, y_n\}_{n=1}^N)}$$

- Note that because $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$,

$$p(\{\mathbf{x}_n, y_n\}_{n=1}^N) = p(y_1, y_2, \dots, y_N) = N \left(\begin{bmatrix} m(\mathbf{x}_1) \\ \vdots \\ m(\mathbf{x}_N) \end{bmatrix}, \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & \cdots & k(\mathbf{x}_1, \mathbf{x}_N) \\ \vdots & \ddots & \vdots \\ k(\mathbf{x}_N, \mathbf{x}_1) & \cdots & k(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix} \right)$$

$$p(\{\mathbf{x}^*, y^*\}, \{\mathbf{x}_n, y_n\}_{n=1}^N) = p(y^*, y_1, y_2, \dots, y_N) = N \left(\begin{bmatrix} m(\mathbf{x}^*) \\ m(\mathbf{x}_1) \\ \vdots \\ m(\mathbf{x}_N) \end{bmatrix}, \begin{bmatrix} k(\mathbf{x}^*, \mathbf{x}^*) & k(\mathbf{x}^*, \mathbf{x}_1) & \cdots & k(\mathbf{x}^*, \mathbf{x}_N) \\ k(\mathbf{x}_1, \mathbf{x}^*) & k(\mathbf{x}_1, \mathbf{x}_1) & \cdots & k(\mathbf{x}_1, \mathbf{x}_N) \\ \vdots & \vdots & \ddots & \vdots \\ k(\mathbf{x}_N, \mathbf{x}^*) & k(\mathbf{x}_N, \mathbf{x}_1) & \cdots & k(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix} \right)$$

GP for Nonlinear Regression (3)

- $p(y^* | \mathbf{x}^*, \{\mathbf{x}_n, y_n\}_{n=1}^N)$ is Gaussian
- By Gaussian conditional,

$$p(y^* | \mathbf{x}^*, \{\mathbf{x}_n, y_n\}_{n=1}^N) = N(\boldsymbol{\mu}^*, \boldsymbol{\Sigma}^*)$$

where

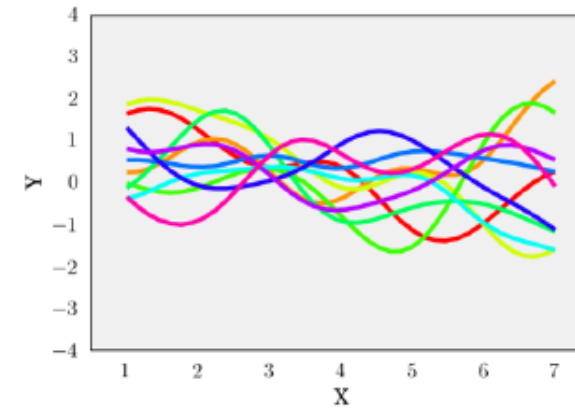
$$\boldsymbol{\mu}^* = m(\mathbf{x}^*) + [k(\mathbf{x}^*, \mathbf{x}_1) \quad \cdots \quad k(\mathbf{x}^*, \mathbf{x}_N)] \cdot \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & \cdots & k(\mathbf{x}_1, \mathbf{x}_N) \\ \vdots & \ddots & \vdots \\ k(\mathbf{x}_N, \mathbf{x}_1) & \cdots & k(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix}^{-1} \left(\begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} - \begin{bmatrix} m(\mathbf{x}_1) \\ \vdots \\ m(\mathbf{x}_N) \end{bmatrix} \right)$$

$$\boldsymbol{\Sigma}^* = k(\mathbf{x}^*, \mathbf{x}^*) - [k(\mathbf{x}^*, \mathbf{x}_1) \quad \cdots \quad k(\mathbf{x}^*, \mathbf{x}_N)] \cdot \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & \cdots & k(\mathbf{x}_1, \mathbf{x}_N) \\ \vdots & \ddots & \vdots \\ k(\mathbf{x}_N, \mathbf{x}_1) & \cdots & k(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix}^{-1} \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}^*) \\ \vdots \\ k(\mathbf{x}_N, \mathbf{x}^*) \end{bmatrix}$$

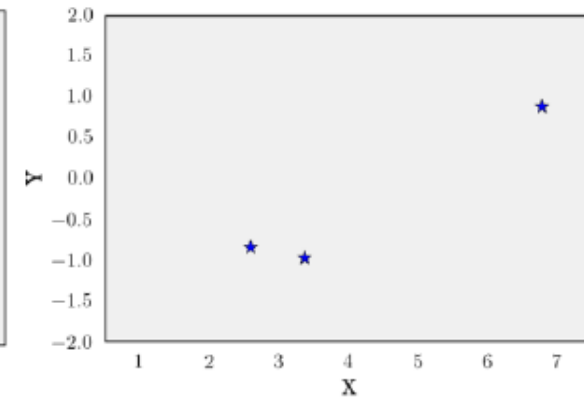
- $\boldsymbol{\mu}^*$: predictive function value at $\mathbf{x}^*$
- $\boldsymbol{\Sigma}^*$: prediction uncertainty

GP for Nonlinear Regression (3)

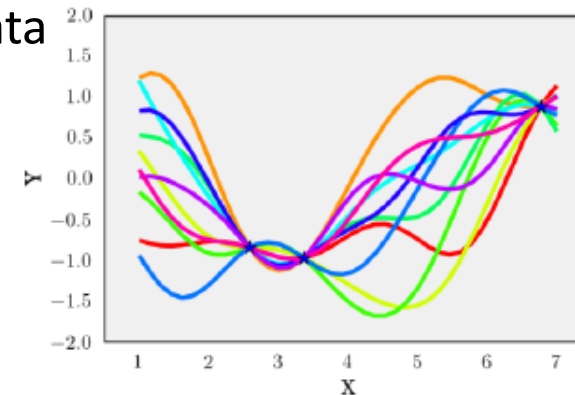
- Example
 - $m(x) = 0$
 - $k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2l^2}\right) + \sigma_v^2 \delta(x - x')$
 - $l = 1$
 - $\sigma_v = 0$: no measurement noise
 - $\sigma_f = 1$
- 'Accurate' prediction around training data points
- 'Large' uncertainty when away from training data



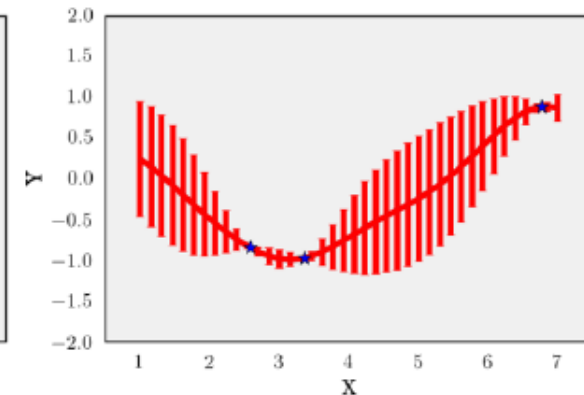
(a)



(b)



(c)



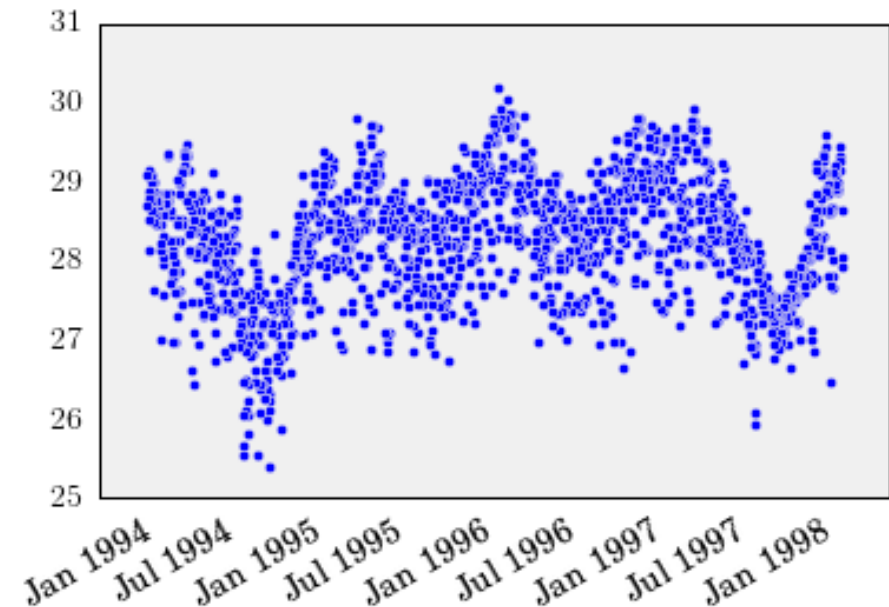
(d)

GP for Nonlinear Regression (4)

- GP-based nonlinear regression is **non-parametric**
 - Complexity proportional to $O(N^3)$, scalable methods needed for large-scale dataset
- GP-based nonlinear regression **explicitly quantifies prediction uncertainty**
- GP-based nonlinear regression is equivalent to a neural network with one hidden layer and **infinitely large number of neurons**
 - Powerful function approximator
- A few mean functions and covariance functions are available for modeling
 - **Combinations of different mean functions and covariance functions possible**

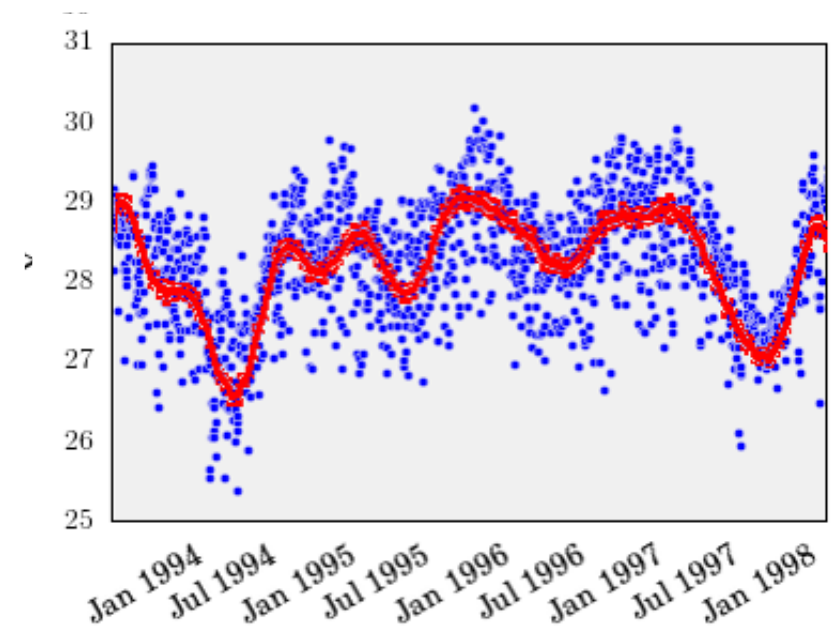
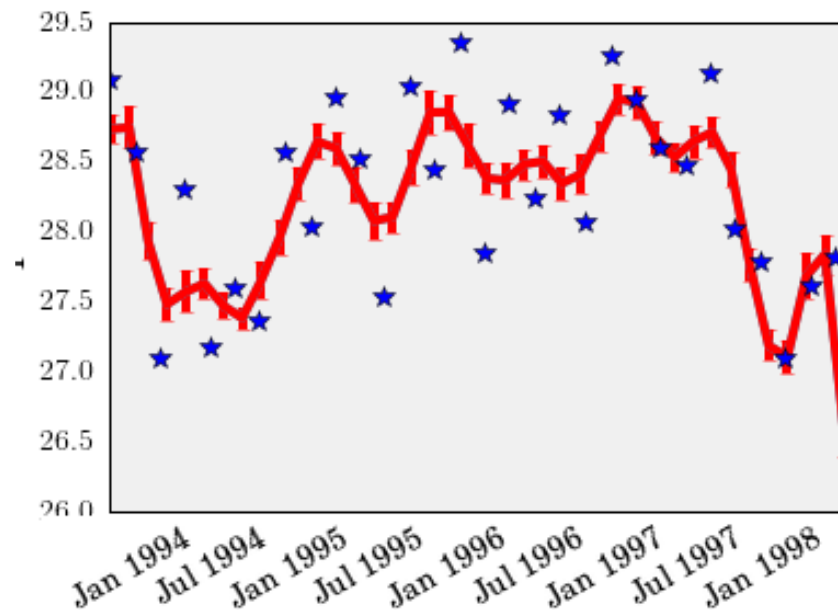
GP for Spatiotemporal Modeling (1)

- Spatiotemporal phenomenon is an event relating to both time and space
- El Nino dataset: 178,080 meteorological measurements across the Pacific
 - Air temperature
 - Relative humidity
 - Surface winds
 - Sea surface temperature
- 25 sensors and span of 5 years (daily records)
- Air temperature at one particular location →



GP for Spatiotemporal Modeling (2)

- Prediction at 40 uniformly spaced time points using 30 measurements
 - Mean function $m(x) = 27$
 - SE covariance function with $l = 100$ days, $\sigma_f = 1$, $\sigma_v = 1$
- Prediction at 200 uniformly spaced time points using 100 measurements
 - Mean function $m(x) = 27$
 - SE covariance function with $l = 100$ days, $\sigma_f = 1$, $\sigma_v = 1$



References

Book Chapters

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